# SYED FAHAD ABBAS KAZMI

**AI Engineer | Full-Stack Software Developer**

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## PROFESSIONAL SUMMARY

AI Engineer and full-stack software developer with 3+ years of professional experience and 1.5+ years shipping LLM systems to production: Retrieval-Augmented Generation (RAG) pipelines, multi-agent workflows, voice AI agents and computer-vision pipelines. Builds end to end in TypeScript (Next.js, Nest.js, React) and Python (FastAPI), on PostgreSQL. Co-Founder & CTO of Traceo, a Chrome extension for bug capture. Designs AI features with deterministic fallbacks so products keep working when a model fails.

## TECHNICAL SKILLS

**AI / ML:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LlamaIndex, Vercel AI SDK, Claude Agent SDK, Embeddings, Vector Databases (pgvector), Prompt Engineering, Agentic Systems, Model Context Protocol (MCP), Computer Vision (Roboflow, Segment Anything Model)

**AI Tooling:** MCP Servers, Claude Code (Skills, Hooks, Sub-agents), n8n Workflow Automation, Vapi Voice AI, Agent Orchestration, Cursor

**Languages:** TypeScript, JavaScript, Python, PHP, SQL, HTML5, CSS3

**Frameworks:** Next.js, React, Nest.js, Node.js, FastAPI, Laravel, Tailwind CSS, Bootstrap, jQuery

**Databases & Infra:** PostgreSQL, MySQL, Supabase, Neon, Redis, BullMQ, Docker, OSRM, Vercel, Oracle Cloud

**Practices:** REST API Design, Third-Party API Integration, Schema Design, Query Optimization, Git, Agile, Responsive Design

## PROFESSIONAL EXPERIENCE

**Co-Founder & CTO** | Traceo **Feb 2026 – Present**

- Co-founded Traceo and lead engineering for its Chrome extension that lets developers, designers and QA teams inspect CSS, capture visual bugs and record sessions, then share them as actionable reports.
- Own extension architecture, front-end engineering in TypeScript and React, and the AI features that explain what broke and why.
- Built Growth Office, a seven-agent AI growth team on the Claude Agent SDK (see Projects).
- Lead product design, the website and launch, and growth measurement.

**AI Engineer** | BXTrack Solutions, Islamabad, Pakistan **Jun 2025 – Present**

- Design and build AI-powered features for production applications, including LLM-driven functionality, RAG pipelines and intelligent automation.
- Built an AI route-planning module for a field-service client that assigns visits by skills, capacity, shifts and time windows; managers approve up to 200 suggestions at once.
- Architect vector database and embedding pipelines that give AI features retrieval-augmented, context-aware answers.
- Develop agentic workflows with MCP servers and custom Claude Code Skills, Hooks and Sub-agents to automate engineering and product tasks.
- Build voice-calling AI agents with Vapi, integrating telephony, LLMs and conversational flows for automated call handling.
- Ship computer-vision pipelines using Roboflow-trained models and the Segment Anything Model (SAM) for object detection and image segmentation.
- Build end-to-end business automations in n8n connecting APIs, databases and AI models.
- Collaborate with cross-functional teams to turn AI capabilities into user-facing features, from prototype to production.
- Evaluate and adopt emerging AI tooling (LangChain, LangGraph, LlamaIndex, Vercel AI SDK) to improve delivery speed and reliability.

**Associate Software Engineer (Contract, Remote)** | BXTrack Solutions **Mar 2025 – Jun 2025**

- Built and maintained Laravel back-end features and REST APIs for client projects, tested with Postman.

**Associate Software Engineer** | BXTrack Solutions, Islamabad, Pakistan **Jun 2023 – Jan 2024**

- Designed, built and maintained full-stack web applications with Laravel and React, from database to UI.
- Developed component-based React front ends and integrated them with Laravel back ends.
- Built and consumed RESTful APIs and integrated third-party services to extend application functionality.
- Optimized performance by refining database queries and reducing page load times across client projects.
- Managed timelines, scope and deliverables, coordinating with stakeholders to ship features on schedule.
- Introduced modern development patterns with the team to improve code quality across projects.

**Laravel Developer (Remote)** | Brainiac Creative Solutions, Sargodha, Pakistan **Jan 2023 – May 2023**

- Architected, developed and maintained scalable web applications using the Laravel framework.
- Collaborated with distributed teams over video calls, chat and email to align on requirements and resolve issues.

**Web Designer (Internship)** | Brainiac Creative Solutions, Sargodha, Pakistan **Oct 2022 – Jan 2023**

- Translated visual concepts into responsive, accessible web interfaces with HTML, CSS, Bootstrap and JavaScript.
- Applied CSS media queries for consistent experiences across devices and screen sizes.

## PROJECTS

**Growth Office: Multi-Agent Growth Team** | Architect & Lead Engineer, Traceo **Sep 2026 – Oct 2026**

*Tech: Claude Agent SDK, Next.js, TypeScript, FastAPI, PostgreSQL, PixiJS | sfak.space/case-studies/growth-office*

- Architected a system of seven AI agents that run GA4 and Search Console standups, technical SEO audits, brand-kit planning, blog drafting and Upwork job scoring for Traceo.
- Gave every LLM step a rule-based fallback and sourced all metrics from real APIs only; publishing stays locked to draft mode behind config.
- Integrated the official Upwork MCP server and a Python (Scrapling/FastAPI) crawler; deployed on Oracle Cloud.

**AI Route Planning & VRP Optimizer** | AI / Backend Engineer, BXTrack Solutions **May 2026 – Sep 2026**

*Tech: Python, FastAPI, NestJS, OSRM, Redis, BullMQ, PostgreSQL | sfak.space/case-studies/ai-route-planning*

- Built the planning engine for a field-service CRM that assigns visits to workers by skills, capacity, shifts, time windows and locked visits.
- Replaced a Google OR-Tools solver with a ~900-line deterministic route builder whose plans planners could predict and trust.
- Cached self-hosted OSRM drive times per road leg and ran planning jobs on a BullMQ queue with a circuit breaker; an LLM only rewrites the reason on each suggestion.

**AI Video Analyzer** | AI Engineer **2025**

*Tech: TypeScript, Next.js, Python, Computer Vision | ai-video-analyzer-v2.vercel.app*

- Built a warehouse-safety platform that detects incorrect parking and waste blocking emergency and fire-brigade access in uploaded footage.
- Delivered two front-end iterations and a Python inference back end covering ingestion, model inference, API design and a results/playback UI.

**Pieces & Bricks Mosaic Studio** | Solo Developer **Oct 2026**

*Tech: TypeScript, React, Vite, MediaPipe, Canvas*

- Built a browser-only proof of concept that turns a photo into a buildable brick mosaic, with in-browser face detection for auto-framing.
- Ported PIL's Lanczos resampling to TypeScript byte-for-byte so output matched a commercial pipeline brick for brick.

**Maqsam Voice Agent** | AI Engineer **2025**

*Tech: TypeScript, Voice AI, LLMs, Telephony | maqsam-voice-agent.vercel.app*

- Built a voice assistant that connects LLMs with telephony so callers interact through an AI-driven conversation; owned the front end, connection state and voice/event handling.

**FloorPlanSegmentor** | AI Engineer **2025**

*Tech: TypeScript, Next.js, SAM, Roboflow | floor-plan-segmentor.vercel.app*

- Built a drag-and-drop tool that extracts structural regions from architectural floor plans with a segmentation pipeline.

**Symptra Policy Assistant** | AI Engineer **2025**

*Tech: TypeScript, Next.js, RAG, LLMs | symptra.vercel.app*

- Built a retrieval-based assistant that answers employee questions about company policies from grounded documents.

**NUML ERP System** | Final Year Project **2025 – Present**

*Tech: Next.js, TypeScript, PostgreSQL, REST APIs*

- Building a university ERP (student, academic and administrative modules) with the official NUML team; contributing full-stack development, schema design and module integration.

**Core42Ops** | Solo Developer **2025**

*Tech: TypeScript, Next.js | core42-ops.vercel.app*

- Built a dashboard for monitoring AI call operations: call volume, completion rate, average duration, recent calls and agent settings.

**Hairstyle-ai** | Solo Developer

*Tech: TypeScript, Next.js, Generative AI | hairstyle-ai-dun.vercel.app*

- Built an AI hairstyle try-on app with 150+ styles on the user's own photo, before/after comparison and a nail-art mode.

**AI Job Scraper** | Solo Developer

*Tech: TypeScript, Web Scraping, LLMs | ai-job-scraper-seven.vercel.app*

- Built an authenticated job-discovery workflow that scrapes listings and uses an LLM to summarize, filter and rank roles.

**BXAssist** | Solo Developer

*Tech: TypeScript, Slack API, LLMs | github.com/1faadi/BXAssist*

- Built a Slack-integrated AI assistant with a policy-chat back end and prompt orchestration layer.

**Next.js AI Chatbot** | Solo Developer

*Tech: Next.js, Vercel AI SDK, TypeScript | github.com/1faadi/nextjs-ai-chatbot*

- Built an LLM chatbot reference architecture with streaming responses and chat persistence, reusable for domain-specific assistants.

**Magic Storybook** | Solo Developer

*Tech: TypeScript, Next.js, Generative AI | ai-gen-storybook.vercel.app*

- Built a generative storybook app that turns a child's photo into the hero of a customized illustrated story.

**BinteShauq** | Solo Developer

*Tech: TypeScript, Next.js | binte-shauq.vercel.app*

- Built an e-commerce storefront for women's wear with catalogue, collections, cart and accounts; owned product, UI and deployment.

## ADDITIONAL PROJECTS

- VercelBot: AI development assistant that automates Vercel deployments, status and project actions (TypeScript).
- Pharmafy: pharmacy management app covering inventory, records and orders (Next.js, PostgreSQL).
- BrandBrain: brand and marketing intelligence tool for marketers (Next.js).
- Cricket Predictor: sports analytics tool with model-driven predictions (TypeScript).
- Ambial: community content platform with authenticated member content (Next.js).
- n8n Render Deploy: configuration for self-hosting n8n workflow automation (Docker).
- Hospital Management System, E-Commerce Platform and Books Portal SPA (Laravel, PHP, MySQL, Vue.js).

## PUBLICATIONS & WRITING

- "I replaced Google OR-Tools with a greedy route builder. Then I deleted every fallback." (Nov 2026), sfak.space/insights
- "I run eleven AI agents on one small server. Here is what they taught me." (Nov 2026), sfak.space/insights
- Regular LinkedIn posts on RAG evaluation, MCP server design, AI cost and agent tooling.

## EDUCATION

**Bachelor of Science in Computer Science** **Feb 2023 – Expected Jan 2027**

National University of Modern Languages (NUML), Islamabad

**Intermediate (HSSC)** **2020 – 2022**

Government College University (GCU), Lahore

**Matriculation (SSC)** **2020**

Sanai School System, Sargodha